One-loop × one-loop interferences

factorisable two-loop families, the $k_1 \cdot k_2$ numerators, and complex conjugation

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Summary

In the real–virtual–virtual contribution at N^{3}LO, the product of two one-loop amplitudes, $\mathcal{A}^{(1)}\,\mathcal{A}^{(1)*}$, contains two loop momenta: k_1 in the amplitude and k_2 in the complex-conjugate amplitude. No propagator depends on both, but the spin and polarisation sums contract Lorentz indices of the two loops, so the numerators contain $k_1 \cdot k_2$. The integrals are therefore not plain products of one-loop integrals, but they belong to a *factorisable* two-loop family. We show:

1. each family has 12 propagators: 3 cut, 4 in k_1, 4 in k_2, and one irreducible numerator for $k_1 \cdot k_2$ (section 2);
2. every integral with powers of $k_1 \cdot k_2$ is a combination of products of one-loop integrals without it (Lorentz covariance, section 4); the reduction of the two-loop family therefore yields masters that are products $M_a(k_1)\,M_b(k_2)$;
3. the reduction never mixes k_1 and k_2 propagators, and its coefficients are real, so the opposite $i0$ prescription of the conjugate loop is applied at the end, $M_a\,M_b^*$; symmetries exchanging k_1 and k_2 must not be used (section 5).

No tensor reduction is needed: the reduction of the two-loop family does it.

1 Setting

One emission: $q(p_1) + \bar{q}(p_2) \to V(q) + g(r_1)$ and crossings, $s = (p_1+p_2)^2$, $q^2 = m^2 = zs$, and the rapidity fixed by u. The cut conditions fix r_1 completely,

$$2\,p_1 \cdot r_1 = s\,\frac{1-uz}{1+u}, \qquad 2\,p_2 \cdot r_1 = s\,\frac{u-z}{1+u}, \tag{1}$$

so every invariant is a function of (s, z, u). The interference

$$\sum_{\text{spins, pols}} \mathcal{A}^{(1)}(k_1)\;\mathcal{A}^{(1)}(k_2)^* \tag{2}$$

is written with the loop momentum k_1 in the amplitude and k_2 in the conjugate. The amplitude loop has Feynman propagators $1/(\ell^2 + i0)$; the conjugate loop has $1/(\ell^2 - i0)$.

Where $k_1 \cdot k_2$ comes from. The open quark line of the amplitude and that of the conjugate join into one Dirac trace, e.g. $\mathrm{Tr}[\,\not{p}_2\,\Gamma(k_1)\,\not{p}_1\,\bar{\Gamma}(k_2)\,]$, and the gluon polarisation sum $-g^{\mu\nu}$ and the leptonic tensor contract a Lorentz index of the amplitude with one of the conjugate. A $\not{k}_1$ from Γ and a $\not{k}_2$ from $\bar{\Gamma}$ in the same trace give $k_1 \cdot k_2$; two pairs give $(k_1 \cdot k_2)^2$, and so on.

2 Counting: why 12 propagators

Integration momenta $\{r_1, k_1, k_2\}$ (the emission and the two loops), external momenta $\{p_1, p_2\}$. The scalar products that involve an integration momentum are

group	scalar products	number
emission	$r_1^2,\ r_1\cdot p_1,\ r_1\cdot p_2$	3
loop 1	$k_1^2,\ k_1\cdot p_1,\ k_1\cdot p_2,\ k_1\cdot r_1$	4
loop 2	$k_2^2,\ k_2\cdot p_1,\ k_2\cdot p_2,\ k_2\cdot r_1$	4
mixed	$k_1\cdot k_2$	1
		12

A family needs 12 linearly independent denominators (or numerators) to express all of them:

- 3 cut propagators: r_1^2, $(p_1+p_2-r_1)^2-m^2$, and the linear rapidity denominator;
- 4 propagators of the box in k_1: $k_1^2,\ (k_1+P_1)^2,\ (k_1+P_2)^2,\ (k_1+P_3)^2$, with P_i the partial sums of the external momenta in one cyclic ordering of the legs p_1, p_2, r_1, q; they are independent in $k_1^2, k_1\cdot p_1, k_1\cdot p_2, k_1\cdot r_1$;
- 4 propagators of the box in k_2, the same for k_2;
- 1 irreducible numerator for $k_1\cdot k_2$, e.g. $(k_1+k_2)^2$: it never appears in a denominator.

$3+4+4+1=12$. Since the cuts fix r_1, one may equally reduce in k_1, k_2 only, at fixed (s,z,u): 9 scalar products, $4+4$ propagators and the same numerator. The two descriptions have the same loop part.

One loop $\times$ tree (RV) has no such numerator: the conjugate has no loop momentum, and the contraction only produces $k_1\cdot p_i$ and $k_1\cdot r_1$, which the box propagators express.

3 Factorisable families

A two-loop family is *factorisable* if its propagators split into two groups depending on disjoint loop momenta: here the k_1-box and the k_2-box. As a graph it is two one-loop graphs with no propagator in common. An integral of the family is

$$I(\vec{a};\vec{b};n)=\int\frac{\mathrm{d}^dk_1}{i\pi^{d/2}}\frac{\mathrm{d}^dk_2}{i\pi^{d/2}}\,\frac{(k_1\cdot k_2)^n}{\prod_j D_j(k_1)^{a_j}\ \prod_j E_j(k_2)^{b_j}}, \tag{3}$$

(D_j the k_1-box propagators, E_j the k_2-box ones, negative a_j, b_j allowed). For $n=0$ it is a product of one-loop integrals,

$$I(\vec{a};\vec{b};0)=J_1(\vec{a})\,J_2(\vec{b}),\qquad J_1(\vec{a})=\int\frac{\mathrm{d}^dk_1}{i\pi^{d/2}}\frac{1}{\prod_j D_j(k_1)^{a_j}}, \tag{4}$$

because the integrand is a product of a function of k_1 and a function of k_2. The only coupling between the two loops is the numerator $(k_1\cdot k_2)^n$.

4 Removing $k_1\cdot k_2$

4.1 Lorentz covariance

Write the k_1-part of the integrand as $f(k_1)=1/\prod_j D_j(k_1)^{a_j}$. It depends on k_1 only through k_1^2 and $k_1\cdot p_i$, with $\{p_i\}=\{p_1,p_2,r_1\}$ (three linearly independent external momenta). Split k_1 into its component in the space spanned by the p_i and the rest:

$$k_1^\mu=k_\parallel^\mu+k_\perp^\mu,\qquad k_\parallel^\mu=\sum_{i,j}p_i^\mu\,(G^{-1})_{ij}\,(k_1\cdot p_j),\qquad G_{ij}=p_i\cdot p_j, \tag{5}$$

and the same for k_2. "Parallel" means: the projection of k_1 onto the three-dimensional subspace spanned by p_1, p_2, r_1; it is the combination of the p_i with the same scalar products as k_1, since contracting with p_l gives $k_\parallel \cdot p_l = \sum_{ij} G_{li}(G^{-1})_{ij}(k_1 \cdot p_j) = k_1 \cdot p_l$. "Perpendicular" means: $k_\perp = k_1 - k_\parallel$ is orthogonal to every p_l, $k_\perp \cdot p_l = 0$. $k_\parallel$ and $k_\perp^2 = k_1^2 - k_\parallel^2$ are functions of the scalar products of the family, and $k_\perp$ lives in the $(d-3)$-dimensional space orthogonal to all p_i. The projector onto that space is

$$g_\perp^{\mu\nu} = g^{\mu\nu} - \sum_{i,j} p_i^\mu \, (G^{-1})_{ij} \, p_j^\nu, \qquad g_\perp^{\mu\nu} p_{l\,\nu} = 0, \qquad g_\perp^{\mu\nu} g_{\perp\,\mu\nu} = g_\perp{}^\mu{}_\mu = d-3. \tag{6}$$

We need the tensor integrals

$$T^\mu = \int d^d k_1 \, f(k_1) \, k_\perp^\mu, \qquad T^{\mu\nu} = \int d^d k_1 \, f(k_1) \, k_\perp^\mu k_\perp^\nu. \tag{7}$$

General form. The integrand of $T^{\mu\nu}$ depends on k_1 only through scalar products, and the integration runs over all of k_1, so $T^{\mu\nu}$ is a Lorentz-covariant function of the p_i: $\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma T^{\rho\sigma}(p) = T^{\mu\nu}(\Lambda p)$. A covariant tensor built from vectors p_i is a combination of products of the metric, of the p_i, and of the Levi-Civita tensor, with coefficients that are functions of the invariants $p_i \cdot p_j$. The Levi-Civita tensor does not enter, for two independent reasons:

1. $T^{\mu\nu}$ is symmetric by construction, while every rank-two structure with one Levi-Civita tensor, such as $\epsilon^{\mu\nu\rho\sigma} p_{1\rho} p_{2\sigma}$, is antisymmetric in $\mu\nu$;

2. f is parity even. A reflection of one direction orthogonal to all p_i (it exists for $d > 3$) leaves the p_i and f unchanged, so $R^\mu{}_\rho R^\nu{}_\sigma T^{\rho\sigma} = T^{\mu\nu}$, while it flips the sign of every structure with one Levi-Civita tensor ($\det R = -1$). This excludes such structures at every rank, T^μ included.

Hence

$$T^\mu = \sum_i C_i \, p_i^\mu, \qquad T^{\mu\nu} = A \, g^{\mu\nu} + \sum_{i,j} B_{ij} \, p_i^\mu p_j^\nu, \qquad B_{ij} = B_{ji}. \tag{8}$$

Transversality. $k_\perp \cdot p_l = 0$ holds before integration, so $T^\mu p_{l\,\mu} = 0$ and $T^{\mu\nu} p_{l\,\nu} = 0$ for every l. For the vector, $\sum_i C_i G_{il} = 0$ for all l, and G is invertible, so $C_i = 0$. For the tensor,

$$A \, p_l^\mu + \sum_{i,j} B_{ij} \, G_{jl} \, p_i^\mu = 0 \quad \Longrightarrow \quad A \, \delta_{il} + (BG)_{il} = 0 \quad \Longrightarrow \quad B = -A \, G^{-1}, \tag{9}$$

since the p_i are linearly independent. Hence $T^{\mu\nu} = A \, g_\perp^{\mu\nu}$: $g_\perp$ is the only combination of g and the p_i orthogonal to all p_i.

Trace. Contracting with $g_{\mu\nu}$ and using $g_\perp{}^\mu{}_\mu = d-3$, $A = \frac{1}{d-3} \int d^d k_1 \, f(k_1) \, k_\perp^2$, an integral of the family since $k_\perp^2 = k_1^2 - k_\parallel^2$ is a function of its scalar products. Hence, inside the integral,

$$k_\perp^\mu \to 0, \qquad k_\perp^\mu k_\perp^\nu \to \frac{k_\perp^2}{d-3} \, g_\perp^{\mu\nu}. \tag{10}$$

The same steps at higher rank give zero for every odd product of $k_\perp$'s and symmetrised products of $g_\perp$ for every even one, e.g. $k_\perp^\mu k_\perp^\nu k_\perp^\rho k_\perp^\sigma \to \frac{(k_\perp^2)^2}{(d-3)(d-1)} \big(g_\perp^{\mu\nu} g_\perp^{\rho\sigma} + g_\perp^{\mu\rho} g_\perp^{\nu\sigma} + g_\perp^{\mu\sigma} g_\perp^{\nu\rho} \big)$.

The same result without the general form. At fixed $k_\parallel$, i.e. at fixed $k_1 \cdot p_i$, the integrand depends on $k_\perp$ only through $k_\perp^2 = k_1^2 - k_\parallel^2$. The integral over $k_\perp$ is therefore invariant under the rotations of the $(d-3)$-dimensional orthogonal space, and $T^{\mu\nu}$, which lives entirely in that space, must be a symmetric invariant tensor of it: a multiple of its metric $g_\perp$.

Levi-Civita tensors in the numerator. The traces with γ_5 can leave Levi-Civita tensors contracted with k_1, e.g. $\epsilon(k_1, p_1, p_2, r_1) = \epsilon_\mu(p_1, p_2, r_1)\, k_\perp^\mu$. They are not part of f: they are external tensors contracted with $k_\perp^\mu$ or $k_\perp^\mu k_\perp^\nu$, and the replacements above apply to them as to k_2. The derivation needs only f, the denominators and the scalar-product numerators, to be parity even, which always holds.

4.2 Rank one

Expanding,

$$k_1 \cdot k_2 = k_\parallel \cdot k_{2\parallel} + k_\parallel \cdot k_{2\perp} + k_\perp \cdot k_{2\parallel} + k_\perp \cdot k_{2\perp} = k_\parallel \cdot k_{2\parallel} + k_\perp \cdot k_{2\perp}, \tag{11}$$

because the cross terms vanish identically: $k_\parallel$ is a combination of the p_i and $k_{2\perp}$ is orthogonal to all of them (and vice versa). The first term is $k_\parallel \cdot k_{2\parallel} = \sum_{ij}(k_1 \cdot p_i)(G^{-1})_{ij}(k_2 \cdot p_j)$; the second is odd in $k_\perp$ and drops out:

$$I(\vec{a}; \vec{b}; 1) = \sum_{i,j}(G^{-1})_{ij} \Big[\int f(k_1)\,(k_1 \cdot p_i) \Big] \Big[\int g(k_2)\,(k_2 \cdot p_j) \Big]. \tag{12}$$

Each bracket is a one-loop integral of the family without $k_1 \cdot k_2$: $k_1 \cdot p_i$ is a combination of box propagators and constants. So $I(\vec{a}; \vec{b}; 1)$ is a sum of products $J_1\, J_2$ with rational coefficients.

4.3 Rank two

$(k_1 \cdot k_2)^2 = (k_\parallel \cdot k_{2\parallel})^2 + 2(k_\parallel \cdot k_{2\parallel})(k_\perp \cdot k_{2\perp}) + (k_\perp \cdot k_{2\perp})^2$. The middle term is odd in $k_\perp$; the last one gives $k_\perp^\mu k_\perp^\nu k_{2\perp\mu} k_{2\perp\nu} \to k_\perp^2\, k_{2\perp}^2/(d-3)$. Hence

$$(k_1 \cdot k_2)^2 \;\to\; (k_\parallel \cdot k_{2\parallel})^2 + \frac{1}{d-3}\, k_\perp^2\, k_{2\perp}^2, \tag{13}$$

and both terms are products of a function of the k_1 scalar products and one of the k_2 scalar products. Any power n works the same way.

4.4 Can $k_1 \cdot k_2$ survive in a master?

No relation is missing: the identities above are Lorentz-invariance identities, and these are linear combinations of the integration-by-parts identities of the family [1]. A complete reduction of the two-loop family therefore contains them: every integral with $n > 0$ reduces to integrals with $n = 0$, which are products. Which integrals are called masters is a choice of basis: an ordering that ranks numerator powers as more complicated (the usual one) never picks an integral with $n > 0$. The masters are then products,

$$\{\text{masters}\} = \{M_a(k_1)\, M_b(k_2)\}, \tag{14}$$

with $\{M_a\}$ the one-loop masters of the k_1-box and $\{M_b\}$ those of the k_2-box: the IBP identities generated by $\partial/\partial k_1 \cdot v$ with $v \in \{k_1, p_i\}$ act on the k_1 factor only and are exactly the one-loop IBPs, so the master basis of the product is the product of the two bases.

Sanity checks.

- The coefficients contain $1/\det G$ and $1/(d-3)$. With $p_1^2 = p_2^2 = r_1^2 = 0$,

$$\det G = 2\,(p_1 \cdot p_2)(p_1 \cdot r_1)(p_2 \cdot r_1) = \frac{s^3}{4}\,\frac{(1-uz)(u-z)}{(1+u)^2}, \tag{15}$$

  non-zero inside the physical region and vanishing on the collinear boundaries $u = z$ and $u = 1/z$, where r_1 is parallel to p_2 or p_1 and the three momenta no longer span three directions. These are the boundaries where the expansion in distributions is needed anyway. $d-3 = 1-2\epsilon$ is harmless.

- In a subsector the k_1-integral may depend on fewer external momenta (a triangle or a bubble); the same argument holds with the Gram matrix of those momenta.
- A direct check, on a family with both boxes in the ordering p_1, p_2, r_1: eight integrals, with $n = 0, 1, 2$, one dot, and in subsectors, computed in two ways, (i) with the rank-one and rank-two formulas above followed by the one-loop reduction of each factor, and (ii) by the reduction of the two-loop family without $k_1 \leftrightarrow k_2$ symmetries (section 5). The two agree term by term as ordered products $M_a(k_1)\, M_b(k_2)$, and no master carries a power of $k_1 \cdot k_2$.

5 Complex conjugation

The k_1 propagators carry $+i0$, the k_2 propagators $-i0$. This is consistent with the reduction:

1. The IBP identities are algebraic: they do not depend on the $i0$ prescription, and their coefficients are rational functions of s, z, u, d with real coefficients.
2. No identity produces a propagator depending on both k_1 and k_2: $\partial/\partial k_1$ acts only on the k_1 propagators and on the numerator $k_1 \cdot k_2$. Every term of the reduction keeps the split (normal k_1 factor) $\times$ (conjugate k_2 factor).

Hence one reduces as if both prescriptions were $+i0$, and evaluates each master as

$$M_a(k_1; +i0)\; M_b(k_2; -i0) = M_a\; M_b^*, \tag{16}$$

with M_b the usual (Feynman) one-loop master at the same real kinematics.

Symmetries. Symmetries acting on k_1 alone (shifts and reflections of the k_1-box) or on k_2 alone preserve the split and may be used. A symmetry that exchanges k_1 and k_2 (possible when both boxes have the same ordering) must not be used: it would identify $M_a\, M_b^*$ with $M_b\, M_a^*$, i.e. move the conjugation to the other factor. The exchange relates instead the two interferences with amplitude and conjugate swapped, $\mathcal{A}_i^{(1)} \mathcal{A}_j^{(1)*} = \left(\mathcal{A}_j^{(1)} \mathcal{A}_i^{(1)*}\right)^*$, which is used when the contributions are summed, not inside the reduction.

An automatic search for sector relations finds this exchange: for two boxes of the same ordering it maps, for instance, (box in k_1)$\times$(bubble in k_2) onto (bubble in k_1)$\times$(box in k_2), so the reduced integral is correct as a number for $+i0$ in both loops but no longer correct once the conjugation is applied to the k_2 factor. The reduction of the family is therefore run with sector relations and sector symmetries switched off; every sector is then reduced on its own, and the masters are the $4 \times 4 = 16$ ordered products. In the check of section 4 the reduction with the exchange allowed agrees with the formulas of section 4 for all eight integrals only after symmetrising each product, $M_a M_b \to \frac{1}{2}(M_a M_b + M_b M_a)$, and as ordered products for one of them; without the exchange all eight agree as ordered products.

6 Families and reductions

- Families: (ordering of the k_1-box) $\times$ (ordering of the k_2-box), 3 orderings each (legs p_1, p_2, r_1 around the box, q fixed, up to reflection), at most 9 families of 12 propagators.
- Two different reductions enter the single-emission virtual corrections: the one-loop boxes (one loop $\times$ tree, NNLO and N^{3}LO) and the factorisable two-loop families of this note (one loop $\times$ one loop).
- Masters: products of one-loop masters, each needed to the order in ϵ required by the product, with the conjugation applied to the k_2 factor. With massless internal lines the triangles reduce to bubbles, so each box has four masters: the box and the three bubbles in the invariants of its two-particle channels and in q^2 (for the ordering p_1, p_2, r_1: $(p_1 + p_2)^2$, $(p_2 + r_1)^2$, q^2); a family has $4 \times 4 = 16$ masters.

References

[1] R. N. Lee, *Group structure of the integration-by-part identities and its application to the reduction of multiloop integrals*, JHEP 07 (2008) 031, arXiv:0804.3008.